MONGODB HARNESS ENGINEERING & MODEL WRANGLING · SF TECH WEEK · OCT 11, 2026

One person + a team of self-improving agents = 10×

Every employee gets specialist agents that improve every week from how people actually work. Raindrop finds where they fail. MongoDB Atlas holds every version and the proof.

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MongoDB Atlas Voyage AI Raindrop Modelence

THE PROBLEM

Company agents ship once and never get better

STATIC

They never learn

The prompt written on launch day is still the prompt a year later. Every rephrase and correction is thrown away.

SILENT

Nobody gives feedback

Thumbs and forms go unused. The real signal is what people do next: rephrase, correct, build on it, copy it.

UNSAFE

Self-editing is risky

An agent that rewrites its own playbook can get worse, leak data, or widen its own access.

THE PROPOSAL

Learn from what people do. Keep only what the data proves.

LEARN · BEHAVIOUR

Every action is a signal

Rephrase, correction, follow-up intent, regenerate and copy are captured automatically. No feedback forms.

FIND · RAINDROP

Know where it fails

Every answer is a Raindrop event. The Query API tells each agent which answers fail, why, and on which version.

PROVE · MONGODB ATLAS

Version every change

Each change is a new playbook version in Atlas, kept only if it beats its parent on unseen tests. Rollback is one update.

Three triggers start an improvement cycle. Five gates decide whether a change is kept.

SOLUTION · THE PRODUCT

Every person gets a team of agents, in their own context

Hand off a job with `@release_agent`. The agent answers with your role, persona and projects, and sees only the documents IAM allows. Higher roles inherit the agents below them.

Model agnostic: local Gemma or Llama, or any cloud model by config. Private data stays on local models.

ROLE	OWN AGENTS	TOTAL
Developer	Build, Verifier, Debug, Release, Code Review	**5**
Product manager	Spec, Prioritization, Research, Launch, Metrics	**5**
Director	Status, Risk, People, Capacity, Incident Review + dev and PM agents	**15**
CTO	Strategy, Board Brief, Architecture, Vendor, Security + all below	**20**

SOLUTION · HOW IT WORKS

One loop, from answer to proven improvement

1 — 2 — 3 — 4 — 5 — 6

MONGODB ATLAS

Answer

`$vectorSearch` with IAM compiled into the pre-filter. Logged as an episode.

RAINDROP

Observe

Each answer is an event; each rephrase, correction or follow-up a signal.

RAINDROP QUERY API

Diagnose

Failures by agent, failure type and playbook version.

IMPROVEMENT CYCLE

Propose

One change, from the week's ledger in Atlas and the cause of failure.

GATES

Prove

Unseen tests, blind side-by-side replay, leak test, 2+ people.

MONGODB ATLAS

Keep or undo

A new version with parent and evidence, or recorded as undone.

↺ The next answer uses the new version. A watchdog compares live Raindrop failure rates and rolls back if it is worse.

SOLUTION · WHEN IT IMPROVES

Three triggers start an improvement cycle

RAINDROP ERROR SPIKE

30%

An agent's Raindrop failure rate crosses 30% over 24 hours, with at least 5 answers. The cycle starts immediately, with a 6-hour cooldown.

SCHEDULE

7 days

Every agent and every person runs a cycle on the last week's decision ledger, even when nothing looks wrong.

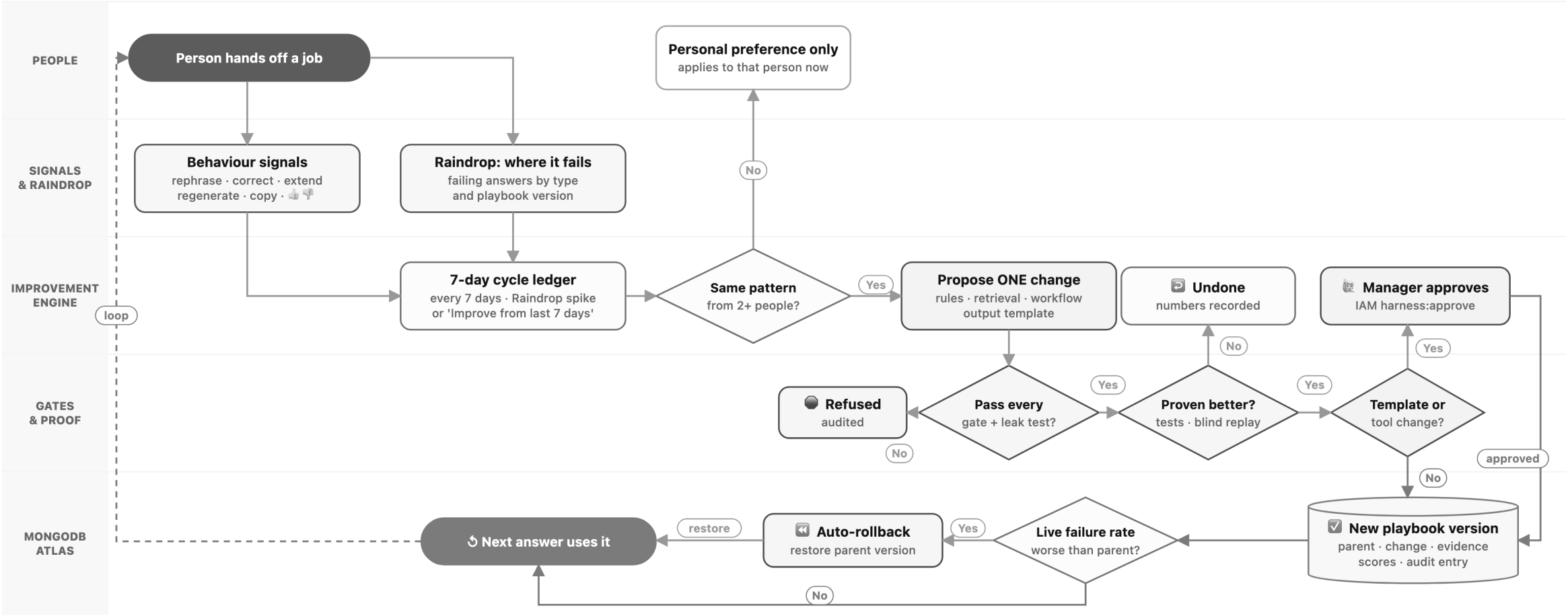
ON REQUEST

On demand

A person clicks "Improve from last 7 days". IAM decides who may: a PM can start a cycle but cannot tune an agent directly.

Whatever starts it, the cycle runs the same steps and passes the same gates.

SOLUTION · END TO END



SOLUTION · MONGODB ATLAS

Atlas is the agents' memory and their proof

Runs on the Atlas Sandbox: Cluster0, MongoDB 8.0, database `omniq_atlas`. Every number in this deck is a query over it.

`$vectorSearch`	4 indexes. IAM policies compile into the pre-filter, so the database enforces access.
`$rankFusion`	Hybrid vector and Atlas Search retrieval, a setting the loop can switch on.
`Voyage via Atlas`	voyage-4 embeddings and rerank-3 through the Atlas Model API key.
`Versioned documents`	Every playbook, persona and context layer, with parent, change, evidence and scores.
`Change streams`	Each step of an improvement cycle streams live to the screen.
`$graphLookup · TTL`	Company ontology with access control. TTL locks keep cycles from colliding.
`Audit chain`	Hash-chained log of every decision, verified end to end.

SOLUTION · RAINDROP

Raindrop sits inside the loop, not beside it

WRITE

Every answer is an event

The Raindrop event id is the Atlas episode id, so the two systems join exactly. Rephrase, correction, regenerate, copy and follow-up intent are sent as signals.

READ

Query API in every cycle

Failing answers per agent, by failure type, and the failure rate of each playbook version. Shown to users as "Where it fails".

ACT

Triggers and rollback

A 30% error spike starts a cycle. After release, the watchdog compares versions and rolls back automatically.

SOLUTION · GUARDRAILS

Agents improve every week and cannot make themselves worse

1

Same pattern from 2+ people?

No → personal preference only

2

Passes every gate and the leak test?

No → refused and audited

3

Better on unseen tests and blind replay?

No → undone and recorded

4

Template or tool change?

Yes → a manager approves

5

Live failures worse than parent?

Yes → automatic rollback

No agent can edit IAM, change who sees what, or skip a gate. Names are masked before any model call.

RESULTS · QUERIED FROM MONGODB ATLAS

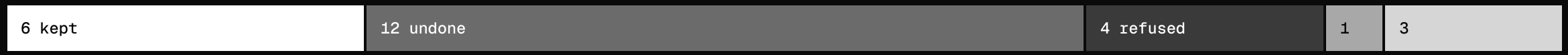
Every change was decided by data

26
automatic playbook decisions

+38
points average gain on kept changes

0
leaks in any kept version

78%
of 90 answers right the first time



6 kept · 12 undone · 4 refused by safety rules · 1 awaiting a manager · 3 other. 83 Raindrop events · 21 improvement cycles · 84 graded tests · e2e 29/29 on the public URL.

RESULTS · BUSINESS VALUE

Each person does the work of a team

OUTPUT

10× per person

Specialist agents for every role, handed a job with one @ mention.

KNOWLEDGE

Compounds weekly

What people correct becomes the agent's playbook, instead of being lost in chat.

QUALITY

Mistakes stop repeating

Raindrop finds the failure. The next version fixes it, or it is undone.

RISK

Safe to deploy

Access enforced by MongoDB, zero leaks, a full audit trail, no model lock-in.

RESULTS · TRY IT

Pick a person and hand off a job

1. Sign in as Diego (Developer). Ask `@release_agent can I ship on friday afternoon`, rephrase it, then ask for the rollback checklist.
2. My agents → Release Agent → Where it fails.
3. Learning → Improve from last 7 days.
4. Impact, then Under the hood → Access and Data layer.

omniq-mv4a4zfo47x-prod.cloud.modelence.app	Live app
…/workflow	Flowcharts
github.com/sandeepvijayarao09/omniq-atlas	Code